MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, K8 OPERATION AND SEQUENCE

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprises the complete requirements for the K8 Operation and Sequence Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10554164	Electronic Component Assembly, K8 Operation and Sequence

PACKAGING DATA SHEET

10554164	Packaging of Circuit Card Assembly, K8 Operation and Sequence

FSC 4931

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554164 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3.3.2 of MIL-C-60898, the requirements of 3.4 shall be met with exceptions as specified in the following sub-paragraphs.

3.3.1.1 Logic "1" output. An output logic “l” shall be greater than +2.5 vdc but less than +4.4 vdc.

3.3.1,2 Logic “O” output. An output logic “O” shall be greater than +0.01 vdc but less than +0.4 vdc.

3,3.1.3 Pin 27 output. The output voltage of pin 27 shall be either +20.5 ± 1.5 vdc or greater than +0.05 vdc but less than +1.2 vdc.

3.4 Electrical performance

3.4.1 Conditions of operation. The requirements of 3.4.2 shall be met with the terminations to the board as shown in Figure 1. A logic “O” input shall be +0.50 ± 0.01 vdc and a logic "1" input shall be +2.80 ± 0.01 vdc.

3.4.2 Sequence and timing. With the inputs applied in the sequence specified in Table I, there shall be the outputs specified in Table I at pin 17 and at pin 27; pin 1 shall be GROUND. An output logic “l” shall be greater than +2.8 vdc but less than +4.4 vdc. An output logic “O” shall be greater than +0.01 vdc but less than +0.4 vdc.

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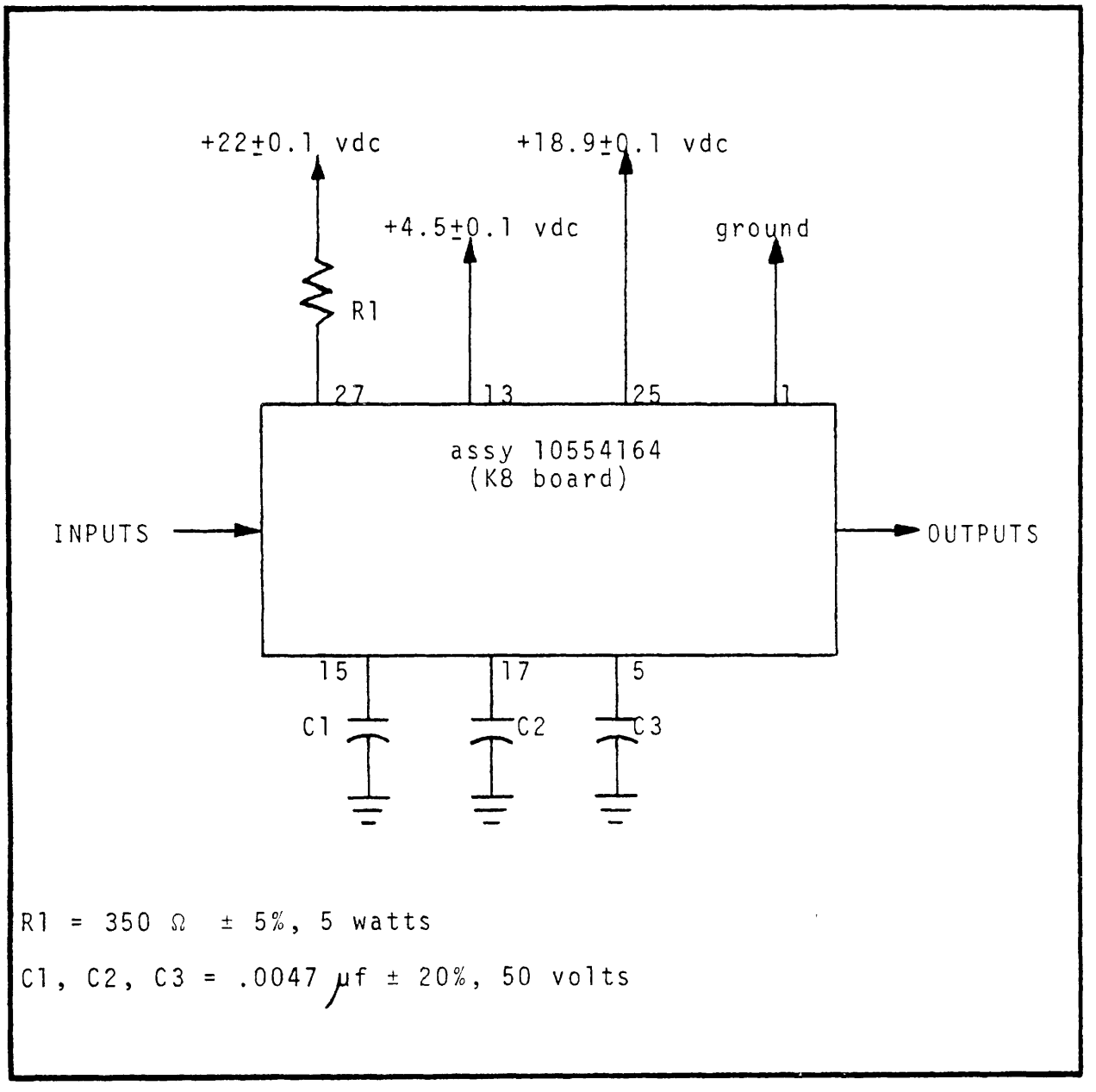


Figure 1

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Table I

Step	Input pins			Output pins	
	3	9	11	17	27
1	0	**	1	0	+21.5 ± 0.5 vdc
2	1	1	1	1*	+0.05 vdc to +1.2 vdc
3	0	1	1	1	+21.5 ± 0.5 vdc
4	0	**	0	0	+21.5 ± 0.5 vdc
5	1	1	0	1*	+21.5 ± 0.5 vdc
6	0	1	0	1	+0.05 vdc to +1.2 vdc
7	0	**	0	0	+21.5 ± 0.5 vdc

* Logic "1" delayed 200 ± 25 milliseconds from the time pin 3 becomes a logic "1".

** Apply a "0" for 1 to 5 seconds and then apply a "1".

3.4.3 Capacitor C3. With the signal shown in Figure 2 applied through 100 ohm ± 10 percent resistor, between pin 25(+) and pin 1, the voltage out between pin 25(+) and pin 1 shall be as shown in Figure 3.

3.4.4 Capacitor C2. With the signal shown in Figure 2 applied through a 100 ohm ±10 percent resistor, between pin 13(+) and pin 1, the voltage out between pin 13(+) and pin 1 shall be as shown in Figure 3.

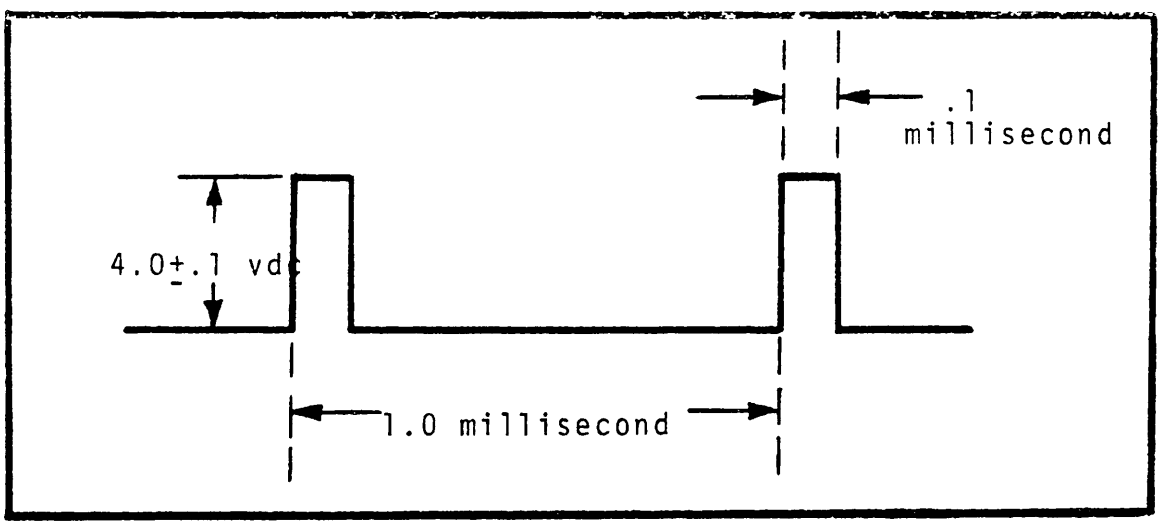


Figure 2

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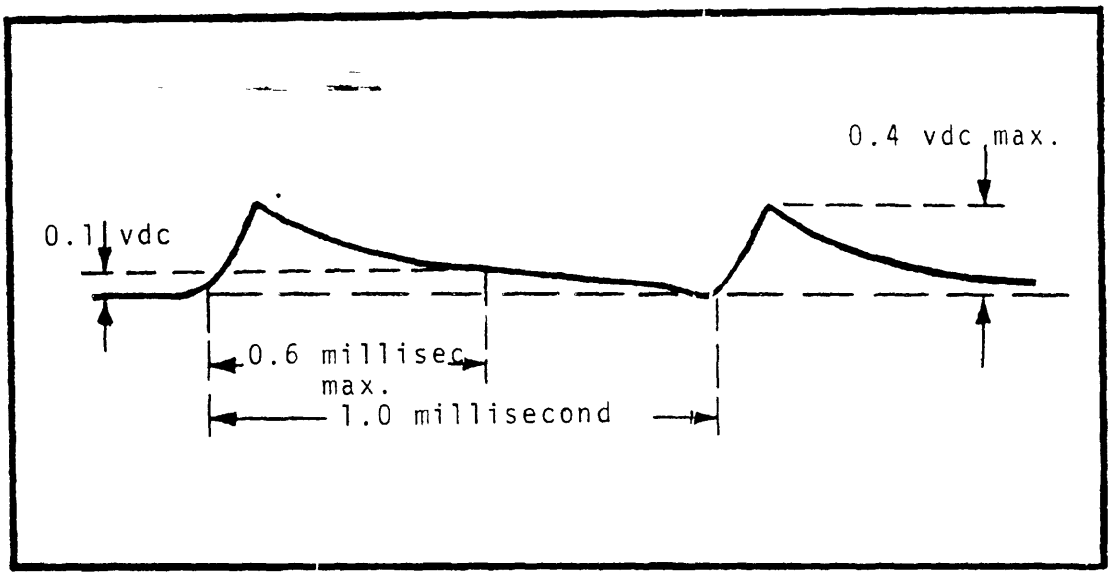


Figure 3

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL.C-60898.

4.3 control test sample-~. As specified in MIL-C-60898.

4.4 Functional tests. The following characteristics shall be 100 percent tested in accordance with the applicable test method.

	Characteristic	Requirement	Procedure
100	Conditions of operation	3.4.1	4.5.2.1
101	Sequence and timing	3.4.2	4.5.2.2
102	Capacitor, C3	3.4.3	4.5,2.3
103	Capacitor, C2	3.4.4	4.5.2.4

4.4.1 Inspection equipment-. As specified in MIL-C-60898 and 4.4.1.1.

4 ,4.1.1 Test equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

Item		Description
1.	POWER SUPPLY	22.0 ± 0.1 vdc, 500ma or greater. Ripple less than 100 mv rms.
2.	POWER SUPPLY	4.5 ± 0.1 vdc, 500ma or greater. Ripple less than 100 mv rms.
3.	POWER SUPPLY	18.9 ± 0.1 vdc, 500ma or greater. Ripple less than 100 mv rms.
4.	POWER SUPPLY	0.50 ± 0,01 vdc, 50ma or greater. Ripple less than 5 mv rms.
5.	POWER SUPPLY	2.80 ± 0.01 vdc, 50ma or greater. Ripple less than 5 mv rms,

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6.	DIGITAL VOLTMETER	With accuracy and resolution of ± 0.2 percent or better from .0090 to 99.0 vdc. Input impedence 10 megohms or greater on all voltage scales.
7.	OSCILLOSCOPE	Voltage and time accuracy and resolution of ±5 percent or better. Rise time 100 nanoseconds or less; sweep periods from 1 microsecond to 1 second per centimeter. Amplitude scales from 0.01 to 10 volts/ centimeter, automatic triggering.
8.	ASSORTED COMPONENTS	One resistor 350 ohm ± 5 percent,5 watt. Three capacitors, 0.0047 uf ± 20 percent, 50 vdc. One resistor 100 ohms ± 10 percent, ½ watt.
9.	PULSE GENERATOR	Frequency - 1 kilohertz minimum. Duty factor - 10 percent minimum. Amplitude - minimum of 4 volts into a 200 ohm load. Pulse rise and fall times - less than 0,2 microseconds.

4.5 Test methods and procedures

4.5.1 Environmental . As specified in MIL-C-60898 except the requirements of 3.3,1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

4.5.2.1 Conditions of operation. Unless otherwise specified, all subsequent tests are to be performed with the conditions of operation specified in 3.4.1. Referring to Figure 1, connect R1, Cl, C2, C3 and ASSORTED COMPONENTS, item 8 of 4.4.1.1 Connect the POWER SUPPLIES, items 1, 2 and 3 of 4.4.1.1, to supply +22.0 ± 0.1 vdc, +4.50 ± 0.01 vdc and +18.9 ± 0.1 vdc, respectively, with their negative terminals tied to GROUND as shown in Figure 1. When required, during subsequent tests, use the POWER SUPPLY, item 4 of 4.4.1.1, positive output as logic “O” and the POWER SUPPLY, item 5 of 4.4.1.1, positive output as logic "1". Connect the negative terminals of these two supplies to GROUND of Figure 1.

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4.5.2.2 Sequence and timing. Referring to Table I, apply logic inputs as shown for pins 3, 9, and 11 for each step of Table I. Connect the OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 17 (+) and pin 1. Connect the OSCILLOSCOPE trigger input between pin 3 and pin 1. Adjust. the OSCILLOSCOPE to trigger when pin 3 goes from a logic "0" to a logic "'l", Using the DVM, verify that the output voltages on pins 17 and 27 conform to the requirements of Table 1 and 3.4.2 for each specified combination of inputs on pins 3, 9, and "Il. With the OSCILLOSCOPE, verify that pin 17 goes from logic "O" to logic "l" within 200 ± 25 milliseconds after pin 3 is changed from a logic "O" to a logic "1", with all other input pins at their required logic level, for steps 2 and 5 of Table I.

4.5.2.3 Capacitor C3. Connect the PULSE GENERATOR, item 9 of 4.4. 1.1, thru the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 404.1.1, between pin 25 (+) and pin 1 per Figure 4. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B vertical input, per Figure 4, and verify the waveform of Figure 3.

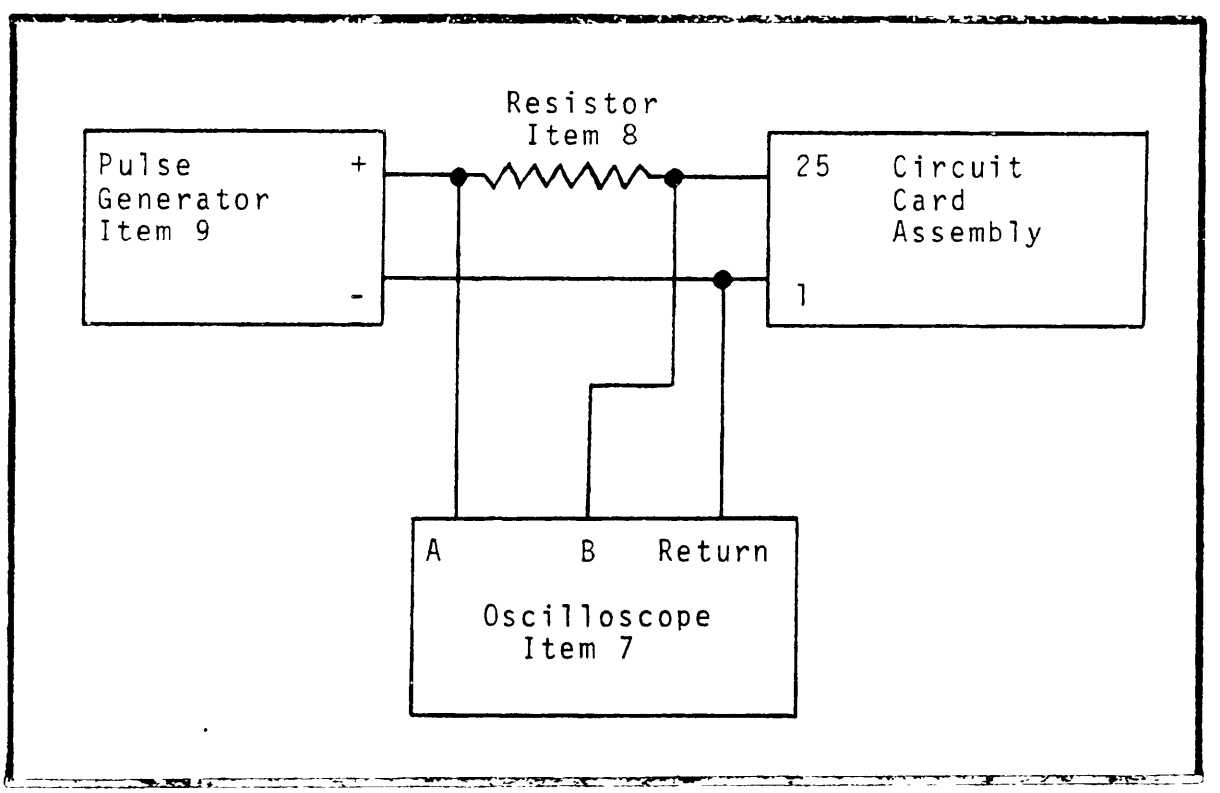


Figure 4

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4.5.2.4 Capacitor C2. Connect the PULSE GENERATOR, item 9 of 4,4.1.1, thru the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 13 (+) and pin 1 per Figure 5. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B vertical input, per Figure 5, and verify the waveform of Figure 3.

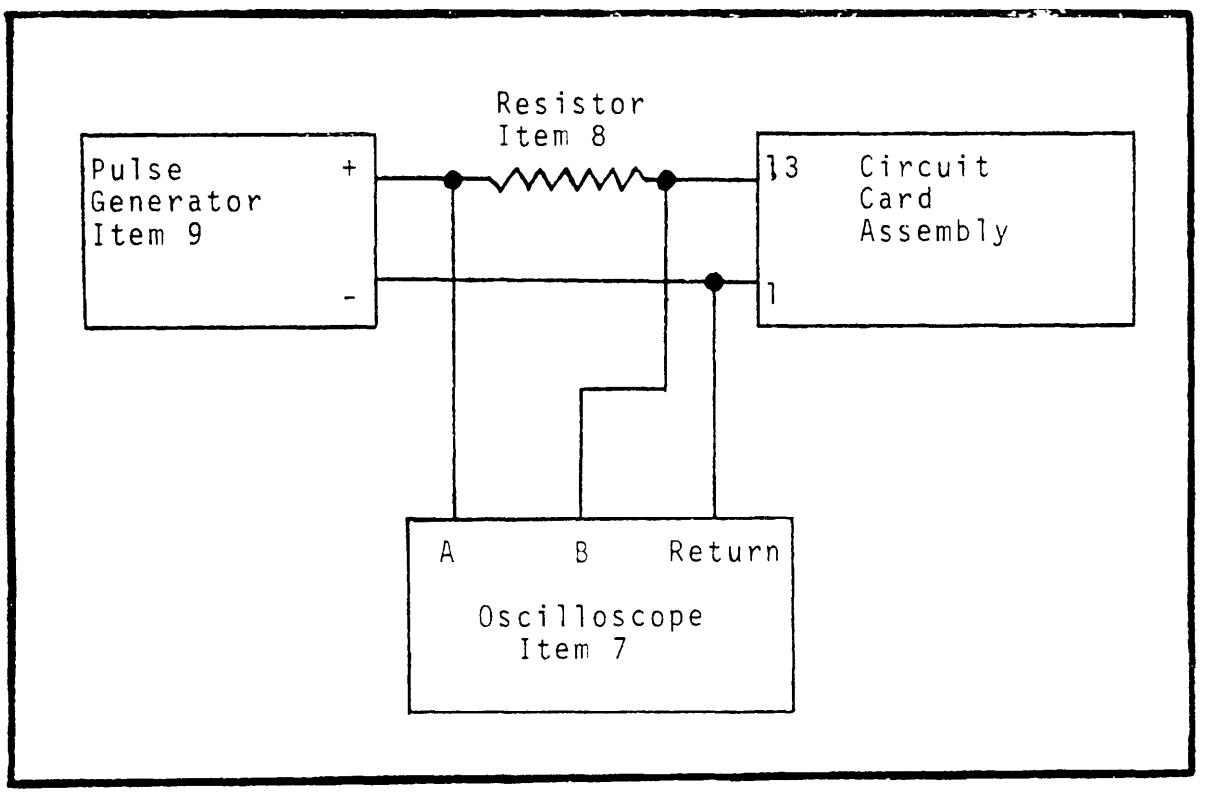


Figure 5

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554164.

6.. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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